DELIVERABLE D3.2

Technical Requirements and Specifications Report

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SEMDR

Smart Energy Management and Demand Response

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Abstract

This document defines the technical requirements and system specifications for the SEMDR (Smart Energy Management and Demand Response) project, an AI-driven energy management platform targeting the Cypriot hospitality sector. As Deliverable D3.2, it establishes the architectural blueprint, hardware and software specifications, and measurable operational requirements that guide the system's design, procurement, and deployment at the pilot site.

The system architecture is organised into four functional layers: a data acquisition layer comprising industrial energy meters, environmental sensors, and edge computing gateways; a secure communication layer based on Wi-Fi 6 mesh networking and encrypted cloud connectivity; a cloud-hosted data management and analytics layer incorporating demand response optimisation algorithms; and an application layer providing hotel staff with real-time monitoring and control via a web-based dashboard. Nine operational requirements (OR1-OR9) are defined, covering data availability, guest comfort thresholds, algorithm execution timing, dashboard responsiveness, non-disruptive deployment, fault tolerance, flexibility market readiness, scalability, and data protection compliance.

The requirements and specifications presented here are informed by the state-of-the-art review (D3.1) and align with the project's scientific and technical objectives as set out in the original proposal. They provide the baseline against which system validation (D3.4) and performance reporting will be conducted. This document also identifies candidate assets for demand response, including HVAC chillers, swimming pool pumps, and refrigeration systems, and outlines initial considerations for their inclusion in the pilot deployment.

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1 Introduction

Cyprus's hospitality sector is a cornerstone of the national economy and a significant consumer of electricity. Heating, ventilation and air conditioning (HVAC) systems alone can account for more than half of a hotel's total energy consumption, with additional loads from refrigeration, water heating, and pool operations. Many hotels still rely on manual or outdated control strategies, resulting in inefficiencies, higher costs, and missed opportunities to support renewable integration.

In parallel, Cyprus's electricity grid is increasingly shaped by high penetration of intermittent renewable sources such as solar and wind. While these resources are abundant, their variability leads to renewable curtailment and volatile energy prices. To address these challenges, the **Smart Energy Management & Demand Response (SEMDR)** project aims to design, deploy, and validate a smart energy management system tailored to hotels.

By integrating **real-time monitoring**, **IoT-based data acquisition**, and **demand response optimisation algorithms**, SEMDR will:

- Reduce peak demand and electricity costs.
- Provide hotels with access to flexibility markets.
- Ensure guest comfort and operational reliability.

The project follows a **technology roadmap** advancing from **TRL 5 to TRL 7**:

- **TRL 5**: Core algorithms validated in simulation and laboratory environments.
- **TRL 6**: Adaptation and testing in a pilot hotel, validating algorithm performance under operational conditions.
- **TRL 7**: Full deployment in a partner hotel, demonstrating robustness across multiple occupancy patterns and seasons.

This deliverable (D3.2) defines the technical requirements and system specifications for the SEMDR platform. It serves as the primary reference document for hardware procurement, software architecture decisions, and system integration activities across WP3,WP4, and WP5. The requirements set out here translate the project's scientific and technical objectives which were established in the original proposal and refined through the state-of-the-art review (D3.1), into concrete, measurable specifications that will guide the system's design, development, and validation.

The document is structured as follows: Section 3 summarises the concept, objectives, and operational requirements derived from the proposal; Section 4 presents the system architecture, component specifications, and hardware/software selection for the pilot deployment. The requirements traceability matrix in Section 5 maps each objective to the system components and work packages responsible for its delivery.

2 Concept and Objectives

2.1 Current State of the Art

The design of the SEMDR system is grounded in a comprehensive review of the current state of demand response, building energy management, and data-driven optimisation in the hospitality sector. This review has been conducted in full in D3.1, and only the key conclusions that have directly shaped the technical requirements in this document are summarised here.

International experience has established that hotels can reliably curtail 10-30% of peak electricity demand through coordinated load management of HVAC systems, swimming pool pumps, lighting, and operational equipment, provided these measures are executed via centralised Building Energy Management Systems (BEMS) with minimal guest impact [1] [2]. HVAC systems typically account for over 50% of a hotel's total electricity consumption and represent the single largest opportunity for demand response [3]. Pool pump filtration cycles constitute a further significant deferrable load whose scheduling can be optimised against dynamic electricity pricing.

Algorithmically, the field has progressed beyond simple rule-based scheduling toward Model Predictive Control (MPC), Dynamic Programming (DP), and machine learning-based approaches - including gradient boosting models used as surrogate models of building thermal dynamics within optimisation frameworks [4] [5]. These methods enable real-time, cost-optimal control that balances energy expenditure against comfort constraints and equipment longevity.

In Cyprus, the energy management landscape in the hospitality sector remains significantly behind global best practice. Most hotels operate with manual or basic automated systems lacking real-time analytics, dynamic optimisation, or demand response capabilities. Cyprus's islanded electricity grid, characterised by growing but variable renewable generation and volatile day-ahead pricing, creates both an urgent need and a favourable environment for DR adoption. The country's Integrated National Energy and Climate Plan (INECP) identifies approximately 50 MW of untapped demand response potential by 2030, with hotels and resorts among the most promising participants [6].

For the complete literature review, including international case studies, a detailed Cyprus gap analysis, and a review of algorithmic approaches, the reader is referred to Deliverable D3.1 - State of the Art Review.

2.2 Formulated Concept

The SEMDR system is conceived as an integrated, IoT-enabled energy monitoring and demand response platform that can be deployed within an operational hotel environment without disrupting guest services or requiring replacement of existing building infrastructure. The system bridges the gap between international best practice in hotel energy management and the current state in Cyprus by introducing real-time data acquisition, cloud-hosted analytics, and algorithmic optimisation capabilities to a sector that has, to date, relied on manual or basic automated energy control.

The system architecture is organised around four functional layers: a data acquisition layer comprising energy meters, environmental sensors, water quality monitors, and edge gateways; a secure communication layer using Wi-Fi mesh networking and encrypted cloud connectivity; a data management and analytics layer hosted on AWS cloud infrastructure for storage, processing, and algorithmic decision-making; and an application layer providing hotel staff with real-time visibility and control through a web-based dashboard.

Two primary demand response use cases have been selected for the pilot deployment at Elias Beach Hotel, Limassol: HVAC chiller optimisation and swimming pool pump scheduling. These were chosen because they represent the two largest flexible electrical loads in a typical hotel, and because their operational constraints - thermal comfort for HVAC, water quality for pools - are well-characterised and measurable. Additional candidate assets, including refrigeration systems and lighting, are being evaluated for potential inclusion during later project phases; refrigeration is of particular interest due to its significant base load and thermal storage characteristics, though its suitability for demand response will depend on findings from the data collection and analysis phase (WP4). Lighting loads, while referenced in the original proposal, have been assessed as offering limited demand response potential in the pilot hotel context, where modern lighting systems already operate at relatively low power levels compared to HVAC and pool systems, and where guest-facing lighting adjustments carry higher comfort risk. The project will document these scope decisions formally in the Use Case Scenarios and KPIs Report (D3.3). In both primary use cases, the system optimises energy consumption against day-ahead electricity prices from the Cyprus flexibility market while maintaining strict safety and comfort boundaries.

2.3 Aims and Objectives

2.3.1 Aim

The SEMDR project aims to design, develop, and validate a smart energy management and demand response system for the Cypriot hospitality sector that reduces electricity costs, supports grid stability, and maintains guest comfort. The project will demonstrate that advanced energy management technologies - including IoT-based real-time monitoring, cloud-hosted data analytics, and algorithmic demand response optimisation - can be effectively deployed within an operational hotel environment, providing a replicable model for the wider adoption of demand-side flexibility across Cyprus's tourism infrastructure and similar Mediterranean markets.

The system is piloted at Elias Beach Hotel in Limassol, targeting the two highest-impact use cases identified in the state-of-the-art review: HVAC chiller optimisation and swimming pool pump scheduling. By integrating real-time energy and environmental data with day-ahead electricity pricing from the Cyprus flexibility market, the SEMDR platform enables automated, cost-optimal load management that responds dynamically to grid conditions while respecting operational constraints and guest comfort thresholds. The project follows a technology readiness roadmap advancing from TRL 5 (algorithm validation in simulation) through TRL 6 (pilot hotel testing) to TRL 7 (full operational deployment), enabling progressive validation under increasingly realistic conditions.

2.3.2 Scientific and Technical Objectives

The specific scientific and technical objectives set in the project proposal to meet the above aim are:

SO1) Implement real-time energy monitoring across key hotel systems: IoT-based monitoring systems will be installed and integrated on essential hotel infrastructure including HVAC chillers, pool pumps, and refridgeration systems, collecting energy consumption and environmental data at 1-minute intervals. Data will be securely transmitted to a cloud-based platform for continuous real-time monitoring.

SO2) Develop demand response algorithms for the hospitality sector: Demand response algorithms will be developed and refined specifically for hotels, enabling predictive load management, non-intrusive load shifting, and peak shaving. The target is a 10% reduction in peak energy consumption while maintaining guest comfort within ±1.5°C of set points.

SO3) Develop and deploy a user-friendly energy management interface: A user-centric dashboard will be designed to allow hotel staff to visualise real-time energy usage, costs, and alerts for abnormal patterns. The system will feature manual override capabilities, enabling rapid response to automated DR events with a response time of under 30 seconds.

TO1) Quantify energy and cost savings: The project will demonstrate a 3-5% reduction in energy consumption of connected assets compared to the pre-installation baseline, without compromising guest comfort. This includes calculating the Return on Investment (ROI), targeting a payback period of less than 24 months.

TO2) Optimise system performance for scalability: The system will be designed to handle high-granularity data from multiple assets without degradation. The architecture will support fully automated demand reduction commands and allow seamless integration of additional systems, such as on-site renewable energy sources.

TO3) Validate system reliability and guest satisfaction: System uptime will be maintained at over 95% throughout the project, and comfort-related complaints will be resolved within 24 hours. Monthly surveys with hotel staff will ensure system expectations are met and guest comfort is preserved.

TO4) Prepare for future energy market participation: A simulation model will be developed to explore how the hotel can participate in future energy flexibility markets, with an expected increase in energy cost savings of 10-15%. The model will also explore the potential of adding energy storage under new market scenarios.

2.3.3 Operational Requirements

For the above technical/scientific objectives to be achieved the following operational requirements have been set:

OR1) Real-time data availability: The monitoring infrastructure must achieve continuous data collection at 1-minute intervals with a minimum uptime of 95%, ensuring that the cloud platform receives uninterrupted energy and environmental data streams from all connected assets.

OR2) Guest comfort within defined thresholds: All demand response actions must maintain indoor temperatures within ±1.5°C of set points and pool water ORP levels above 650 mV. The system must automatically revert to safe default states if comfort or water quality boundaries are at risk of being breached.

OR3) Algorithm execution within scheduling cycle: Optimisation algorithms must complete their scheduling computations and issue control commands within each 30-minute scheduling cycle, ensuring that demand response decisions are applied in time for the next pricing interval.

OR4) Dashboard responsiveness and usability: The web-based dashboard must render real-time data with no more than 60 seconds of latency, support manual override of any automated DR action within 30 seconds, and be usable by hotel operations staff without specialist technical training.

OR5) Non-disruptive deployment: The system must be installable within an operational hotel without requiring shutdown of existing building systems, structural modification, or disruption to guest services. All hardware must interface with existing electrical infrastructure using standard industrial protocols (Modbus RTU over RS-485).

OR6) Fault tolerance and safe defaults: In the event of sensor failure, network interruption, or cloud service unavailability, all controlled equipment must default to its normal operating state. No single point of failure shall leave equipment in an unsafe or energy-wasteful condition.

OR7) Flexibility market readiness: The system must be capable of ingesting day-ahead electricity price signals from the Cyprus flexibility market and using them as inputs to the optimisation algorithms, enabling automated, price-responsive demand response scheduling.

OR8) Scalability without re-engineering: The system architecture must support the addition of new monitored assets, sensor types, or hotel properties without requiring changes to the core software platform, cloud infrastructure, or communication protocols.

OR9) Data protection compliance: All data transmission must be encrypted (MQTT over TLS, VPN tunnels), access to the cloud platform must be role-controlled, and any personal data collected must comply with GDPR requirements.

3 System Concept Review

3.1 Architecture and Components

The SEMDR system is designed around **four interlinked layers**, each responsible for different aspects of monitoring, optimisation, and control:

1. **Data Acquisition Layer:** This layer forms the physical foundation of the system. It comprises energy meters, environmental sensors, water quality monitors, and edge computing devices deployed on-site at the hotel. These components collect granular, real-time data on electricity consumption, equipment operating status, indoor environmental conditions, and pool water quality. Edge gateways perform local data aggregation and preprocessing before forwarding data to the cloud, ensuring resilience in the event of intermittent connectivity.
2. **Communication Layer:** This layer provides the secure, reliable network infrastructure that connects on-site devices to each other and to the cloud platform. It encompasses the local Wi-Fi mesh network, VLAN segmentation for traffic isolation, VPN tunnels for encrypted remote access, and the MQTT/HTTPS protocols used for cloud data transmission. The communication layer is designed to operate continuously within the hotel's existing network environment without interfering with guest or operational IT services.
3. **Data Management and Analytics Layer:** This layer receives, stores, processes, and analyses the data streams generated by the acquisition layer. Hosted on cloud infrastructure, it includes a relational database for structured, time-indexed storage of sensor readings and energy data, object storage for backups and large datasets, and serverless compute functions for real-time data routing and transformation. The analytics component houses the demand response optimisation algorithms, which use the stored data and external inputs such as day-ahead electricity prices to generate optimal equipment schedules.
4. **Application Layer:** This layer provides the human-facing interface to the system. A web-based dashboard gives hotel staff real-time visibility of energy consumption, system status, active optimisation schedules, cost savings, and alerts. It also supports manual override of any automated demand response action. Open APIs at this layer enable future integration with external platforms, including grid operator systems, third-party energy services, and the hotel's existing building management system.

3.1.1 Data Acquisition Layer

3.1.1.1 Sensors and Meters

Energy-consuming assets such as HVAC chillers, air-handling units, pool pumps, water heaters, and refrigeration units are equipped with meters or sensors that form the primary data sources for the SEMDR platform. These monitoring and control capabilities fulfil the SCADA (Supervisory Control and Data Acquisition) function described in the project proposal, implemented through industrial-grade Modbus RTU communication and IoT edge gateways rather than a traditional centralised SCADA architecture. This distributed approach was selected to maximise scalability

and minimise deployment complexity in operational hotel environments. These devices collect electricity consumption (active power, reactive power, voltage, current, power factor), operational status, and environmental variables such as room temperature, humidity, and occupancy at 1-minute intervals. The high-frequency data collection is essential for establishing accurate consumption baselines and enabling the optimisation algorithms to make informed scheduling decisions.

In the pilot implementation, the consortium has procured multi-function energy meters (Selec MFM376-C-CE-ROHS) for three-phase loads including the central HVAC chillers and pool pump motors. These meters provide comprehensive power quality measurements over Modbus RTU, enabling both real-time monitoring and historical trend analysis. Shelly Plus 1 PM smart relays are specified for single-phase loads, providing both switching control and power monitoring in a compact form factor. Indoor temperature and humidity are monitored using Shelly H&T Gen 3 wireless sensors, which provide the environmental feedback necessary for HVAC comfort verification.

Figure 1 - Selec MFM376-C-CE-ROHS Energy Meter

3.1.1.2 Edge Controllers and Sensors

Three Raspberry Pi 4 Model B (8 GB) kits act as edge gateways, providing the critical bridge between the on-site sensor network and the cloud platform. Each Pi is equipped with two isolated USB-to-RS-485 FTDI adapters to interface with Modbus energy meters and equipment controllers. The use of isolated adapters protects both the Pi and the connected metering equipment from electrical noise and ground loops common in industrial environments.

The edge gateways perform several key functions: polling Modbus registers from connected meters at the required 1-minute interval, buffering data locally in the event of temporary network or cloud connectivity loss, executing control commands received from the cloud optimisation layer, and transmitting aggregated data to AWS IoT Core. For pool pump control, the Raspberry Pi issues 5V relay signals to switch pump operation on or off according to the optimised schedule. An EcO Start + Connect sensor monitors pool water quality parameters including ORP (oxidation-reduction potential) and pH, providing the water quality feedback that the pool pump scheduling algorithm requires to ensure safe operating boundaries are maintained.

Figure 2 - Raspberry Pi 4 Model B (8 GB)

3.1.1.3 IoT Device Integration

Where a hotel has an existing Building Management System (BMS), SEMDR is designed to read data non-intrusively via Modbus RTU over RS-485, enabling monitoring and data acquisition without requiring modification to the existing control infrastructure. This approach satisfies OR-5 (non-disruptive deployment) by overlaying the SEMDR monitoring and optimisation capabilities onto the hotel's installed systems rather than replacing them. The 1-minute polling interval provides sufficient temporal resolution for both real-time dashboarding and algorithmic optimisation while remaining within the communication bandwidth constraints of the RS-485 bus.

Figure 3 - Modbus RTU over RS-485

3.1.2 Communication Layer

3.1.2.1 Site Network

All on-site devices connect to a dedicated Ruijie Wi-Fi 6 mesh network, deployed specifically for the SEMDR system to ensure reliable, low-latency communication without placing additional load on the hotel's guest or operational Wi-Fi infrastructure. A Ruijie EG105GW-X gateway serves as the root access point and network controller, providing VLAN segmentation to isolate SEMDR traffic from other network activity, QoS prioritisation for sensor data streams, and VPN support for secure remote administration.

Three Reyee RAP2260(G) indoor access points act as mesh repeaters, extending coverage throughout the hotel's mechanical plant rooms, lobby areas, and service corridors where the HVAC chillers, energy meters, and environmental sensors are located. A weather-proof Reyee

RAP6262 outdoor access point provides dedicated coverage for the pool area, ensuring reliable connectivity to the pool pump controllers and water quality sensors. Raspberry Pi edge gateways connect via Ethernet where cabling is available, or via Wi-Fi where it is not, with automatic failover between interfaces where both are provisioned.

Figure 4 - Reyee RAP2260(G) Indoor Access Point

Figure 5 - Reyee RAP6262 Outdoor Access Point

3.1.2.2 Router and Uplink

The mesh network connects to the hotel's existing internet uplink through a dedicated router, which handles outbound communication with the AWS cloud platform. All cloud-bound traffic is transmitted over encrypted channels, with VPN tunnels providing an additional layer of security for both data transmission and remote system administration. The network architecture is designed to operate independently of the hotel's primary IT infrastructure, so that any connectivity issues on the SEMDR network do not affect guest services, and vice versa.

3.1.2.3 Cloud Connectivity

Device data is forwarded from the edge gateways to AWS using secure MQTT over TLS for real-time telemetry streams and HTTPS for configuration and control messages. MQTT was selected for its lightweight protocol overhead and native support for publish-subscribe messaging patterns, which are well-suited to IoT sensor data where multiple consumers (database, analytics, dashboard) need to receive the same data stream. AWS IoT Core serves as the cloud-

side MQTT broker, handling device authentication, message routing, and integration with downstream processing services. OpenVPN on the Ruijie gateway provides a secure management channel for remote system administration, firmware updates, and troubleshooting.

3.1.3 Data Management and Analytics Layer

3.1.3.1 Data Ingestion

AWS IoT Core receives all incoming device telemetry and applies IoT Rules to route data streams to the appropriate storage and processing services. Structured time-series data such as energy consumption readings, environmental measurements, water quality parameters, and equipment status is routed into Amazon RDS (PostgreSQL) for high-resolution, indexed storage that supports the fast queries required by both the optimisation algorithms and the real-time dashboard. Data is indexed by timestamp, device identifier, and asset type, enabling efficient retrieval of historical baselines and trend analysis. Amazon S3 provides cost-effective long-term backup storage and serves as the repository for large datasets such as raw Modbus register dumps and algorithm training data.

3.1.3.2 Processing

Real-time data processing is handled by a combination of AWS IoT Core rules and AWS Lambda functions. IoT Core rules perform initial data validation, unit conversion, and routing at ingestion time, filtering malformed or duplicate readings before they reach the database. Lambda functions handle event-driven processing tasks such as anomaly detection (e.g., unexpected power spikes or sensor dropouts), alert generation, and the transformation of raw sensor data into the derived metrics displayed on the dashboard. This serverless architecture scales automatically with data volume and eliminates the need to provision and manage dedicated processing servers.

3.1.3.3 Analytics and Decision Engine

The analytics and decision engine is the core intelligence of the SEMDR system, responsible for generating optimised demand response schedules for the hotel's controllable assets. The engine receives as inputs the real-time and historical data from the PostgreSQL database, day-ahead electricity prices from the Cyprus flexibility market, weather forecasts, and equipment operating constraints. It produces as outputs the optimised operating schedules for HVAC chillers and pool pumps, which are transmitted back through the communication layer to the edge gateways for execution.

The algorithmic approach to be determined during the implementation phase (WP4) will be selected based on the characteristics of each controllable load. Candidate approaches identified in the state-of-the-art review include rule-based heuristic scheduling for loads with simple operating constraints, and Model Predictive Control or Dynamic Programming with machine learning-based forward models for loads with complex thermal dynamics. The

scheduling engine operates on a 30-minute cycle via a task scheduler, ensuring that control decisions are updated in alignment with the electricity market pricing intervals.

3.1.4 Application Layer - User Interface and Integration

3.1.4.1 Dashboard

A web-based dashboard provides hotel operations staff with real-time visibility of the SEMDR system's status and performance. The interface displays current and historical energy consumption by asset, active demand response schedules and their projected cost savings, environmental conditions (indoor temperature, pool water quality), equipment operating status, and system alerts for anomalous conditions or constraint violations. The dashboard is designed to satisfy OR-4 (dashboard responsiveness and usability), rendering data with no more than 60 seconds of latency and supporting manual override of any automated demand response action within 30 seconds.

3.1.4.2 Interoperability

Open APIs are provided at the application layer to support integration with external platforms and future system extensions. These include interfaces for receiving day-ahead electricity price signals from grid operators or flexibility market platforms, integration with third-party weather forecasting services that provide inputs to the optimisation algorithms, and connectivity with the hotel's existing BMS for read access to additional data sources not directly monitored by SEMDR hardware. The API architecture follows RESTful conventions and is documented to facilitate future replication of the SEMDR system at additional hotel properties, supporting the scalability objective (TO-2, OR-8).

3.2 System Architecture

Figure 1 illustrates the end-to-end SEMDR system architecture, showing the data flow from on-premises hardware through the site network to the cloud platform and external services. The architecture is organised into four zones: (i) The **on-premises zone** (left) contains the physical hotel assets and their associated monitoring and control hardware. Three subsystems are shown: the HVAC chillers, which are monitored via Selec energy meters and controlled through the hotel's existing BMS control box; the refrigeration units, which are monitored via their existing controllers and Selec meters as a candidate asset for demand response evaluation; and the pool pumps, which are monitored via Selec consumption meters and controlled directly by the Raspberry Pi edge gateways through relay signals. An EcO Start water quality sensor and its gateway provide ORP and pH data from the pool. A Shelly H&T sensor monitors lobby temperature and humidity. Each subsystem connects to a Raspberry Pi gateway via USB-to-RS-485 adapters; (ii) The **site network zone** (centre) shows the Ruijie Wi-Fi 6 mesh network comprising the EG105GW-X root gateway router, three indoor Reyee RAP2260(G) access points, and one outdoor Reyee RAP6262 access point for pool area coverage. This network carries all sensor data from the edge gateways to the uplink router, which provides the secure connection to the cloud; (iii) The **cloud zone** (right) shows the AWS infrastructure: IoT Core receives device data via MQTT, IoT Rules route data to Amazon RDS (PostgreSQL) for structured storage, Amazon S3 for backup, and Lambda functions for event-driven processing. The AI Decision Maker module houses the demand response optimisation algorithms. Amazon Grafana provides internal monitoring and analytics dashboards.

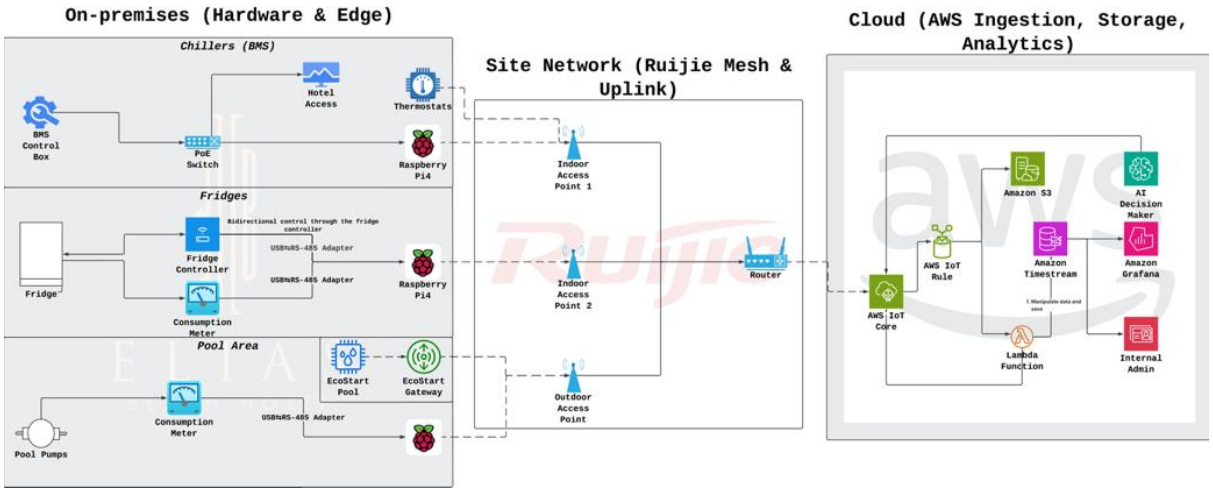

Figure 6 - SEMDR System Architecture

3.3 Hardware and Software Specification Table

The following tables consolidate the hardware and software components specified for the SEMDR system. These specifications define the baseline configuration for the pilot deployment, and serve as the reference for procurement, installation, and system integration activities across WP3, WP4, and WP5. The hardware table lists all physical devices deployed on-site, while the software table lists the platforms, frameworks, and services that comprise the system's data pipeline, analytics engine, and user interface.

3.3.1 Hardware Specification Table

Component	Model / Specification	Purpose	Quantity
3-Phase Energy Meter	Selec MFM376 C CE ROHS	Monitoring of HVAC chillers, pool pumps and main meter supply	3
Smart Relay	Shelly Plus 1 PM	Control/monitoring of single-phase loads	1
Thermostat Sensor	Shelly H&T Gen 3	Indoor temperature and humidity monitoring	1
IoT Gateway	Raspberry Pi 4 (8GB) + FTDI RS485 adapters	Edge gateway, local storage, IoT Core integration	3
Pool Quality Sensor	EcO Start + Connect	Monitoring chlorine & pH	1
Root Gateway Router	Ruijie EG105GW-X	VLANs, VPN, AP root control	1
Indoor APs	Ruijie RAP2260(G)	Wi-Fi 6 mesh coverage (indoor)	3
Outdoor AP	Ruijie RAP6262	Wi-Fi coverage for pool area	1

Table 1 - Hardware Specification Table

3.3.2 Software Specification Table

Component	Technology	Purpose
Edge Software	Python (Modbus libraries, AWS SDK)	Meter polling, local data buffering, relay control, cloud data transmission
IoT Message Broker	AWS IoT Core	Receives device telemetry via MQTT, routes data to storage and processing services
Database	Amazon RDS (PostgreSQL)	Structured storage of time-series energy, environmental, and water quality data
Cloud Storage	Amazon S3	Long-term backup and large dataset storage
Backend & Processing	Django (Python) on AWS EC2, with Celery task scheduler.	Application server, API handling, algorithm orchestration, and scheduled optimisation execution
Frontend Dashboard	Vue.js web application	Real-time energy visualisation, system status, alerts, and manual override controls
Security	OpenVPN, MQTT over TLS	Encrypted remote access and secure device-to-cloud communication

Table 2 - Software Specification Table

4 Requirements Traceability

The following matrix maps the project's scientific and technical objectives to the operational requirements, system components, and work packages responsible for their delivery. This traceability ensures that every objective defined in the project proposal has a corresponding implementation pathway and validation mechanism within the SEMDR system architecture.

Objective	Description	Operational Req.	System Components	WP
SO1	Real-time energy monitoring	OR1, OR5, OR6, OR9	Selec meters, Shelly sensors, RPi gateways, AWS IoT Core, RDS	WP3, WP4
SO2	Demand response algorithms	OR2, OR3, OR7	Analytics engine, Celery scheduler, AWS Lambda, day-ahead price feed	WP4, WP5
SO3	User-friendly dashboard	OR4	Vue.js frontend, Django API, Amazon Grafana	WP5
TO1	Quantify energy/cost savings	OR1, OR2, OR7	Full monitoring stack, analytics engine, baseline comparison module	WP3, WP4, WP5
TO2	System scalability	OR8	Modular architecture, RESTful APIs, cloud auto-scaling	WP5
TO3	System reliability and guest satisfaction	OR1, OR2, OR6	Shelly H&T sensors, safe default logic, uptime monitoring	WP3, WP5
TO4	Future energy market participation	OR7	Day-ahead price API, simulation model, flexibility market interface	WP4, WP5

Table 3 - Requirements Traceability Matrix

5 Preliminary Data Model

At this stage of the project (M3), the data model is preliminary and will be refined as data collection progresses under WP4. The following outlines the high-level data entities and their relationships as currently envisaged for the PostgreSQL database hosted on Amazon RDS. The detailed schema, including field-level specifications, indexing strategy, and data retention policies, will be documented in Deliverable D4.1 (Data Collection and Analysis Report).

The core data entities are as follows. The Assets table stores metadata for each monitored device (chillers, pool pumps, water heaters, refrigeration units), including asset type, location, rated capacity, and communication parameters. The Telemetry table stores time-indexed sensor readings at 1-minute resolution, with fields for timestamp, asset identifier, active power (kW), reactive power (kVAR), voltage, current, power factor, and equipment operating status. The Environmental table stores temperature, humidity, and occupancy readings from Shelly H&T sensors and any future environmental monitoring devices. The Water Quality table stores ORP and pH readings from the EcO Start pool sensor. The Schedules table stores demand response schedules generated by the optimisation engine, including the target asset, scheduled action, start time, end time, and the electricity price signal that triggered the schedule. The Electricity Prices table stores day-ahead pricing data ingested from the Cyprus flexibility market API. All tables are indexed by timestamp and asset/device identifier to support the fast queries required by both the optimisation algorithms and the real-time dashboard.

6 Validation and Testing Approach

While formal acceptance test criteria and validation protocols will be defined in the companion deliverables D3.3 (Use Case Scenarios and KPIs Report, M6) and D3.4 (System Validation and Performance Report, M12), the following provides a brief overview of the intended validation approach to demonstrate how compliance with the requirements defined in this document will be assessed.

Each operational requirement (OR1-OR9) will be validated through a combination of automated monitoring and manual inspection. OR1 (data availability) will be assessed by calculating uptime percentages from the telemetry database over rolling 30-day windows. OR2 (guest comfort) will be validated by comparing indoor temperature logs against the ±1.5°C threshold and pool ORP readings against the 650 mV minimum. OR3 (algorithm execution) will be verified by logging the computation time of each scheduling cycle and confirming completion within the 30-minute window. OR4 (dashboard responsiveness) will be tested through manual interaction timing and automated latency measurement. OR5 (non-disruptive deployment) will be confirmed through hotel operations staff feedback during and after installation. OR6 (fault tolerance) will be validated through deliberate fault injection tests, verifying that equipment reverts to safe default states. OR7 (flexibility market readiness) will be demonstrated by successfully ingesting and responding to day-ahead price signals. OR8 (scalability) will be assessed by adding simulated additional assets to the platform and confirming no degradation. OR9 (data protection) will be confirmed through security audit and penetration testing, as detailed in the Risk Assessment (D1.2).

A structured cybersecurity risk assessment is provided in D1.2 (Risk Assessment Report), which identifies data security and privacy threats and defines corresponding mitigation measures including VPN/TLS encryption, role-based access controls, and periodic security reviews.

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